PAPER_1375_GIBBS_PARADOX

Daniel T. Murphy

PAPER_1375 — UQFF Resolution of the Gibbs Mixing Paradox (CLOSED — EXACT Identity)

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: B — Foundational (CLOSED)
Date: June 16, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — Identity via F_U = 1 global ledger normalization
Calculator surface: `calculate_paradox({"paradox": "gibbs_paradox"})`
Closure helper: `_l96_uqff_axiom_gibbs_paradox_closure()`

The Paradox

For two distinguishable ideal gases of N particles each mixing at constant T, V, the entropy of mixing is:

``
ΔS_mix = 2 N k_B ln 2
``

For two samples of the **same** gas, ΔS_mix = 0 by physical reasoning (no observable change). The discontinuity at the "identical" limit — a finite jump from 2 N k_B ln 2 to 0 as particles become indistinguishable — is the Gibbs paradox.

Classical resolution invokes N! in the partition function via ad-hoc indistinguishability. UQFF resolves this structurally.

UQFF Closed Identity

``
F_U = 1 (global ledger normalization, PAPER_646)
``

Identical particles share the same Universal Aether (UA) mode at the F_U = 1 ledger. Two N-particle samples of the same species are not "two systems" in the UQFF accounting — they are one F_U = 1 distribution with 2N occupation. Mixing produces **no new UA-mode partition**.

``
ΔS_mix^identical_UQFF = 0 EXACT
ΔS_mix^distinguishable = 2 N k_B ln 2 (recovered when species differ)
``

The discontinuity disappears because identical-vs-distinguishable is a **ledger-occupancy question**, not a phase-space-counting question. The F_U = 1 normalization is the natural Gibbs N! correction at the level of the vacuum ledger.

Physical Interpretation

- **Identical particles** occupy a single UA mode at the $F_U = 1$ ledger. The 2N occupation does not produce 2 modes; it produces 1 mode at higher occupation. No mixing entropy.
- **Distinguishable particles** occupy distinct UA modes (different SCm coupling signatures). Mixing produces a genuine multi-mode partition with $\Delta S = 2 N k_B \ln 2$.
- The "paradox" was a classical-statistical-mechanics artifact of double-counting phase space without the $F_U = 1$ constraint.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "gibbs_paradox"})["value"]
```

Field	Value
`delta_S_mixing_identical_particles_per_N_kB_UQFF_EXACT`	**0.0 EXACT**
`delta_S_mixing_distinguishable_per_N_kB_ln_2`	0.6931 (= ln 2)
`F_U_eq_1_global_normalization_makes_identical_share_UA_mode`	1.0
`discontinuity_resolved_via_F_U_eq_1_ledger`	True

C++ Reference Implementation

```cpp
class GibbsParadoxUQFF {
public:
static double deltaS_identical_per_NkB_UQFF() {
return 0.0;
}
static double deltaS_distinguishable_per_NkB() {
return std::log(2.0);
}
static double F_U_global_normalization() {
return 1.0;
}
};
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Statistical Mechanics solve the Gibbs paradox via different methods. SM uses Boltzmann's N! correction in the partition function (combinatorial / phase-space). UQFF derives:

- **ΔS_identical = 0 EXACT** via the F_U = 1 global ledger normalization at the UA-mode level.
- **ΔS_distinguishable = 2 N k_B ln 2** recovered when species occupy distinct UA modes.
- **Zero free parameters.** The resolution is structural, not parametric.

Reference

- UQFF foundational papers: PAPER_646 (F_U = 1 normalization), PAPER_1051 (two-component ρ aether).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_gibbs_paradox_closure`
- Dispatch: `PARADOX_TO_CLOSURE["gibbs_paradox"]`, `["gibbs_mixing"]`
